Supplemental Material for "Exact nonequilibrium steady state of collective group-flip dynamics on quenched interaction neighborhoods"

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S1. NOTATION, CONVENTIONS AND CORRESPONDENCE

Objects. $G = (V, E)$ is a connected graph with $|V| = n$ and graph distance d, which is symmetric. Ball $B_i = B_i(r) = \{j : d(i, j) \le r\}$, $\beta_i = |B_i|$, and β when all balls have the same size; coverage $q = \beta/n$; residual $R_i = V \setminus B_i$. Conditions: **R1** $\beta_i > n/2$ for all i; **R2** $\mathrm{diam}(R_i) \le r$ for all i. Dynamics: each node fires at rate one; a fire at i moves all nodes of state σ_i inside B_i to a target state drawn uniformly among the other $Q - 1$ states ($Q = 2$: the other state). Observables: stationary majority fraction M, magnetization m with $|m| = 2M - 1$; number K of updates between consecutive rotations (changes of the majority label) and cycle time T; containment observables c_G, $\tilde{c}$ as in the main text.

Graphs used. The ring $C_n(1, 2)$ (links to first and second neighbors on a cycle) and the two-dimensional lattice with first- and second-neighbor links along each axis are the 2-section graphs of the uniform hypergrids whose hyperedges are three consecutive nodes along an axis. The group-flip dynamics on a hypergraph depends on it only through its 2-section graph, since $B_i(r)$ is the graph-metric ball of radius r there. All statements below are therefore stated for graphs.

Status of statements. *Theorem, Proposition, Lemma, Corollary*: complete proof given. *Claim*: proof sketch only; not used as a hypothesis of any result. *Conjecture*: numerical evidence only. *Heuristic*: structural statement without proof. *Protocol*: measurement or statistical design. *Note*: methodological record.

Correspondence with the main text. Main-text Theorem 1 is Proposition S4.3; Corollary 1 is Corollary S4.4; Proposition 1 is Proposition S4.5; Lemmas 1–2 are Lemmas S4.7 and S4.10 (the Markov structure used in the main text is Lemma S4.8); Theorem 2 is Theorem S4.13; Corollary 2 combines Theorem S4.15 and Corollary S4.16; Proposition 2 is Proposition S4.18; Proposition 3 is Propositions S5.6–S5.7; Conjecture 1 is Conjecture S8.1.

S2. WELL-MIXED LIMIT AND MULTISTATE VERSION

Lemma S2.1 (Count-chain reduction)**.** *If $B_i = V$ for all i, the state-count vector $x = (x_1, \ldots, x_Q)$, $\sum_k x_k = n$, is a Markov chain with transitions $x \to x + x_k(e_l - e_k)$ at rate $x_k/(Q - 1)$ for each ordered pair (k, l), $k \ne l$.*

Proof. A fire at a node of state k (total rate x_k) moves the entire class k to the target l, chosen with probability $1/(Q - 1)$. The rates depend on the configuration only through x. □

Lemma S2.2 (Pure death of live states)**.** *The number L_t of states with at least one node is a pure-death chain on $\{1, \ldots, Q\}$ with $L \to L - 1$ at rate $n(L - 1)/(Q - 1)$; $L = 1$ is absorbing for L.*

Proof. A fire from live class k decreases L if and only if the target is another live class, which has probability $(L - 1)/(Q - 1)$; summing over live k gives rate $n(L - 1)/(Q - 1)$. A target that is an absent state only relabels the firing class. □

Theorem S2.3 (First-consensus time)**.** *If L_0 states are initially live,*

$$\mathbb{E}[T_{\mathrm{cons}}] = \frac{Q - 1}{n} \sum_{\ell=2}^{L_0} \frac{1}{\ell - 1} = \frac{Q - 1}{n} H_{L_0 - 1};$$

in particular $\mathbb{E}[T_{\mathrm{cons}}] = 1/n$ for $Q = 2$.

Proof. Sum the mean holding times of Lemma S2.2. □

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Theorem S2.4 (Minority advantage, $Q = 2$)**.** *From $(a, n-a)$, the class of size a wins the first consensus with probability $1 - a/n$.*

Proof. The first fire decides: if it is in the class of size a (probability a/n), that class moves into the other state and loses. □

Theorem S2.5 (Three states, one absent)**.** *Let $Q = 3$, states 1 and 2 live with sizes $a, b > 0$, $a + b = n$, and state 3 absent. With $D = a^2 + 3ab + b^2$, the probabilities that state k wins the first consensus are $P_1 = b(a + b)/D$, $P_2 = a(a + b)/D$, $P_3 = ab/D$.*

Proof. Let F be the probability that state 1 wins when states 1 and 2 are live with sizes a and b, G the same with the roles of a and b exchanged, and U the probability that state 1 wins when it is absent and the live classes have sizes a and b. First-step analysis gives

$$\begin{aligned}(a+b)F &= a\big[\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}U\big] + b\big[\tfrac{1}{2}\cdot 1 + \tfrac{1}{2}F\big],\\ (a+b)U &= a\big[\tfrac{1}{2}F + \tfrac{1}{2}\cdot 0\big] + b\big[\tfrac{1}{2}G + \tfrac{1}{2}\cdot 0\big],\\ (a+b)G &= b\big[\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}U\big] + a\big[\tfrac{1}{2}\cdot 1 + \tfrac{1}{2}G\big].\end{aligned}$$

The first and third equations give $F = (aU + b)/(2a + b)$ and $G = (bU + a)/(2b + a)$. Substituting into the second, with $t = a + b$, $p = ab$ and $(2a + b)(2b + a) = 2t^2 + p$, gives $U = p/(t^2 + p) = ab/D$; back-substitution gives F and G. □

Theorem S2.6 (General Q)**.** *Consensus states are absorbing in the well-mixed limit and every other state is transient; the winning probabilities are the unique solution of $(I - P_{TT})x = P_{TA}x_A$, where P_{TT} is the substochastic transient block, invertible by transience. No closed form is claimed for $Q \geq 4$.*

S3. BALL GEOMETRY

Throughout, $\mathrm{diam}(S) = \max_{u,v\in S} d(u, v)$ is measured with the distance d of the whole graph G (not of the subgraph induced by S), and $\mathrm{diam}(\emptyset) = 0$.

Proposition S3.1 (Ball sizes without wrap-around)**.** *On the d-dimensional lattice with first- and second-neighbor links along each axis, the cost of an offset $o \in \mathbb{Z}$ along an axis is $\lceil |o|/2 \rceil$, and as long as no ball wraps around, $|B_r| = \sum_{c_1+\cdots+c_d\leq r} \prod_k w(c_k)$ with $w(0) = 1$, $w(c) = 4$ for $c \geq 1$. Hence $|B_r| = 4r + 1$ in one dimension and $8r^2 + 1$ in two.*

Proof. A single step changes one coordinate by ± 1 or ± 2. Reaching an offset o along one axis needs at least $\lceil |o|/2 \rceil$ steps, and this number suffices (steps of ± 2, plus one step of ± 1 if $|o|$ is odd). Steps along different axes are independent, so the distance is the sum of the axis costs. The offsets of cost $c \geq 1$ along an axis are $\pm(2c - 1), \pm 2c$, four in total, and of cost 0 only $o = 0$; counting offset vectors of total cost at most r gives the convolution formula. In 2D, $(1 + 4r) + \sum_{c=1}^{r} 4\,(1 + 4(r - c)) = 8r^2 + 1$. □

Proposition S3.2 (Ring: distance, balls, R1, R2)**.** *On $C_n(1, 2)$ with nodes labelled by $\mathbb{Z}_n$, let $\delta(i, j) = \min(|i-j|, n-|i-j|)$ be the circular lattice distance. Then (a) $d(i, j) = \lceil \delta(i, j)/2 \rceil$; (b) if $4r + 1 < n$, then $B_i = \{i - 2r, \ldots, i + 2r\}$ (an arc of $\beta = 4r + 1$ nodes), and otherwise $B_i = V$, so $\beta = \min(n, 4r + 1)$; (c) R1 holds iff $2\beta > n$; (d) under R1, R2 holds iff $\beta = n$ or $3\beta \geq 2n - 1$; when $\beta = 4r + 1 < n$ this is equivalent to $n \leq 6r + 2$.*

Proof. (a) A walk of s steps changes the label by at most $2s$ in absolute value (in $\mathbb{Z}$), so reaching j from i requires $2s \geq \delta$, and $s = \lceil \delta/2 \rceil$ steps in the shorter direction suffice. (b) By (a), $j \in B_i$ iff $\delta(i, j) \leq 2r$; if $4r + 1 < n$ these are the $4r + 1$ distinct labels $i - 2r, \ldots, i + 2r$, otherwise every label satisfies $\delta \leq \lfloor n/2 \rfloor \leq 2r$. (c) All balls have size β. (d) If $\beta = n$, residuals are empty. Otherwise $R_i = \{i + 2r + 1, \ldots, i + n - 2r - 1\}$ is an arc of $\ell = n - \beta$ consecutive labels. For two labels of an arc of ℓ consecutive labels the circular distance is at most $\min(\ell - 1, \lfloor n/2 \rfloor)$, with equality for the two end points. Under R1, $\ell < n/2$, so the maximum is $\ell - 1$ and by (a) $\mathrm{diam}(R_i) = \lceil (\ell - 1)/2 \rceil$. R2 is $\lceil (n - \beta - 1)/2 \rceil \leq r$, i.e., $n - \beta - 1 \leq 2r$; with $r = (\beta - 1)/4$ this is $3\beta \geq 2n - 1$, and with $\beta = 4r + 1$ it is $n \leq 6r + 2$. □

Proposition S3.3 (Coverage radius)**.** *For fixed $0 < q < 1$ and d, the smallest radius with $|B_r| \geq qn$ in the no-wrap regime satisfies $r_{\min} = \Theta(n^{1/d})$.*

Proof. By Proposition S3.1, $|B_r|$ is a polynomial in r of degree d with leading coefficient $4^d/d!$, strictly increasing in r; hence $|B_r| \asymp r^d$ and $|B_r| \geq qn$ first holds at $r \asymp n^{1/d}$. □

Claim S3.4 (Wrapped regime; sketch only). *For fixed q on d-dimensional tori, $|B_r|/n$ converges to the rescaled volume of the corresponding diamond with $O(n^{-1/d})$ corrections. Checked numerically only; not used below.*

Claim S3.5 (Sparse configuration model; sketch only). *If $(k-1)^{2r+1} = o(n)$, then $|B_r|$ in the configuration model is asymptotically the non-backtracking tree volume. Not used below; it motivates the truncated-ball design of Protocol S7.1.*

S4. COMPLETE PROOFS OF THE EXACT RESULTS

Logical structure. Section S4 A constructs the process and proves a general time-average lemma (Lemma S4.1) used by every later statement. Section S4 B contains the results that need only R1. Section S4 C defines the epoch chain and proves that it is a Markov chain (Lemma S4.8) and that its states are self-contained (Lemma S4.7). Section S4 D proves the affine closure, Section S4 E the main theorem by a martingale renewal–reward argument that does not assume independent cycles, and Sections S4 F–S4 G the cycle law and the unequal-ball result. Every statement in this section is proved in full.

A. Construction and time averages

State space and update maps. $\Omega = \{0,1\}^V$. For $i \in V$ define $T_i : \Omega \to \Omega$ by $(T_i\sigma)_j = 1 - \sigma_j$ if $j \in B_i$ and $\sigma_j = \sigma_i$, and $(T_i\sigma)_j = \sigma_j$ otherwise. Let $\theta : \Omega \to \Omega$ be the global state exchange $(\theta\sigma)_j = 1 - \sigma_j$. Then $T_i \circ \theta = \theta \circ T_i$, because T_i is defined only through the relation $\sigma_j = \sigma_i$. For $\sigma \in \Omega$ let $\mathcal{M}(\sigma)$ and $\mathcal{A}(\sigma)$ be the majority and minority classes when they are different in size, and $M(\sigma) = \max(|\sigma^{-1}(0)|, |\sigma^{-1}(1)|)/n$.

Construction. Let X_0 be an arbitrary Ω-valued random variable, $(U_k)_{k\geq 1}$ i.i.d. uniform on V, and $(G_k)_{k\geq 1}$ i.i.d. exponential with mean $1/n$, all independent. Put $S_0 = 0$, $S_k = G_1 + \cdots + G_k$, $Y_0 = X_0$, $Y_k = T_{U_k}(Y_{k-1})$, and $X_t = Y_k$ for $S_k \leq t < S_{k+1}$. By the superposition property of Poisson processes (independent rate-one clocks at the n nodes are equivalent to one rate-n clock with uniform marks), (X_t) is the continuous-time Markov chain with generator $Lf(\sigma) = \sum_{i\in V}[f(T_i\sigma) - f(\sigma)]$, which is the group-flip dynamics. (Y_k) is its jump chain (including null jumps) with transition matrix $P = n^{-1}\sum_i T_i$, and $L = n(P - I)$, so a probability vector is stationary for (X_t) iff it is stationary for (Y_k). Let $\mathcal{F}_k = \sigma(X_0, U_1, \ldots, U_k, G_1, \ldots, G_k)$.

A martingale strong law. We use the following standard fact. *If $(D_k)_{k\geq 1}$ is a martingale difference sequence with $\sup_k \mathbb{E}[D_k^2] < \infty$, then $N^{-1}\sum_{k\leq N} D_k \to 0$ almost surely.* Proof: $Z_N = \sum_{k\leq N} D_k/k$ is a martingale with $\mathbb{E}[Z_N^2] = \sum_{k\leq N}\mathbb{E}[D_k^2]/k^2$ bounded, hence it converges almost surely, and Kronecker's lemma gives the claim.

Lemma S4.1 (Time averages). *Almost surely the jump chain (Y_k) enters a closed communicating class $\mathcal{C}$ of P after finitely many steps; let $\mu_{\mathcal{C}}$ be the unique stationary law of P restricted to $\mathcal{C}$. Then for every $f : \Omega \to \mathbb{R}$, almost surely,*

$$\lim_{N\to\infty} \frac{1}{N}\sum_{k=0}^{N-1} f(Y_k) = \lim_{t\to\infty} \frac{1}{t}\int_0^t f(X_s)\,ds = \mu_{\mathcal{C}}(f). \tag{S1}$$

Proof. Ω is finite, so the chain leaves the set of transient states after finitely many steps almost surely and then stays in one closed class, which is irreducible; the ergodic theorem for irreducible finite Markov chains gives the first limit. For the time average, write $\int_0^{S_N} f(X_s)\,ds = \sum_{k=0}^{N-1} f(Y_k)G_{k+1}$. Since G_{k+1} is independent of $\mathcal{F}_k$ and Y_k is $\mathcal{F}_k$-measurable, $D_k = f(Y_k)(G_{k+1} - 1/n)$ is a martingale difference sequence with $\mathbb{E}[D_k^2] \leq \|f\|_\infty^2/n^2$; hence $N^{-1}\int_0^{S_N} f\,ds - n^{-1}N^{-1}\sum_{k<N} f(Y_k) \to 0$, and with $f \equiv 1$, $S_N/N \to 1/n$. For $S_N \leq t < S_{N+1}$, $|\int_0^t f - \int_0^{S_N} f| \leq \|f\|_\infty G_{N+1}$, and $G_{N+1}/N \to 0$ almost surely because $\sum_N P(G_N > \varepsilon N) < \infty$ for every $\varepsilon > 0$ (Borel–Cantelli). Dividing by $t \in [S_N, S_{N+1})$ gives the second limit. □

Remark S4.2 (Definition of $\mathbb{E}[M]$). We define the stationary majority fraction as the almost-sure long-run time average of $M(X_t)$. By Lemma S4.1 it exists and equals $\mu_{\mathcal{C}}(M)$ for the closed class $\mathcal{C}$ that is entered; a priori it may depend on $\mathcal{C}$, hence on the initial law. Under R1∧R2 Theorem S4.13 shows that it does not.

B. Results valid under R1

Proposition S4.3 (Rotation identity; equal balls not required). *Let $Q = 2$ and $\beta_i > n/2$ for every i. Let σ have a strict majority $\mathcal{M}$ and minority $\mathcal{A}$. (a) If $i \in \mathcal{M}$, then $T_i\sigma$ has a strict majority, and its minority is $\mathcal{M} \setminus B_i$, which carries the old majority state: the majority label changes (rotation). (b) If $i \in \mathcal{A}$, then $T_i\sigma$ has minority $\mathcal{A} \setminus B_i$ in*

the same state as before (possibly empty): no rotation. (c) If σ is a tie, then $T_i\sigma$ has a strict majority for every i. Consequently ties are visited at most at time 0, and the instantaneous rotation rate is $|\mathcal{M}| = nM(\sigma)$.

Proof. (a) T_i moves exactly $\mathcal{M} \cap B_i$. The class of the new state is $\mathcal{A} \cup (\mathcal{M} \cap B_i)$ and that of the old state is $\mathcal{M} \setminus B_i$; since $|\mathcal{M} \cap B_i| = \beta_i - |\mathcal{A} \cap B_i|$,

$$|\mathcal{A}| + |\mathcal{M} \cap B_i| - |\mathcal{M} \setminus B_i| = |\mathcal{A}| + 2|\mathcal{M} \cap B_i| - |\mathcal{M}| = 2\beta_i - n + 2|\mathcal{A} \setminus B_i| > 0.$$

(b) T_i moves $\mathcal{A} \cap B_i$ into the majority state; the remaining minority is $\mathcal{A} \setminus B_i$ and the majority only grows. (c) With $|\mathcal{M}| = |\mathcal{A}|$ the computation in (a) applies to either class. By (a) and (b), strict-majority configurations are mapped to strict-majority configurations or to consensus, and consensus ($\mathcal{A} = \emptyset$) is a strict majority. □

Corollary S4.4 (Static–dynamic identity)**.** *Assume R1 and let $\mathcal{C}$ be the closed class entered by (Y_k). Almost surely: (a) the fraction of the first N fires that are rotations converges to $\mu_{\mathcal{C}}(M)$, which is the time average of M; (b) the number of rotations in $[0, t]$ divided by t converges to $n\mu_{\mathcal{C}}(M)$; (c) the average number of fires per rotation converges to $1/\mu_{\mathcal{C}}(M)$.*

Proof. For $k \geq 1$, Y_k has a strict majority (Proposition S4.3(c)), and the $(k+1)$st fire is a rotation iff $U_{k+1} \in \mathcal{M}(Y_k)$. Since U_{k+1} is uniform and independent of $\mathcal{F}_k$, $D_k = \mathbf{1}\{U_{k+1} \in \mathcal{M}(Y_k)\} - M(Y_k)$ is a bounded martingale difference sequence, so the rotation fraction has the same limit as $N^{-1}\sum_k M(Y_k)$, which is $\mu_{\mathcal{C}}(M)$ by Lemma S4.1. (b) follows from (a) and $N(t)/t \to n$, where $N(t)$ is the number of fires in $[0, t]$. (c) is the reciprocal of (a), which is positive since $M \geq 1/2$. □

Proposition S4.5 (One-way erasure transitions)**.** *Assume R1. Let x have nonempty minority $\mathcal{A}$, and let a fire of $j \in \mathcal{A}$ with $\mathcal{A} \subseteq B_j$ map x to the consensus configuration y. Then $T_k(y) \neq x$ for every k. If μ is a stationary law with $\mu(x) > 0$, then μ is not reversible for L.*

Proof. If y is consensus, $T_k(y)$ differs from y exactly on B_k, so the nonconsensus state of $T_k(y)$ occupies B_k, whereas in x it occupies $\mathcal{A}$, and $|\mathcal{A}| < n/2 < \beta_k$ by R1. Reversibility of μ means $\mu(x)L(x, y) = \mu(y)L(y, x)$ for all $x \neq y$; here $\mu(x)L(x, y) \geq \mu(x) > 0$ while $L(y, x) = 0$. □

Remark S4.6. By the argument of Corollary S4.4, the long-run rate of minority fires per unit time is $n(1 - \mu_{\mathcal{C}}(M))$. Under R1 and R2 every minority fire is an erasure (Lemma S4.7), so the long-run rate of one-way transitions is $n(1 - \mathbb{E}[M])$, positive whenever $\mathbb{E}[M] < 1$. In Region II only erasing minority fires are one-way in the sense above; irreversibility holds as soon as one such transition has positive stationary rate, which is the case for the six odd Region-II cases of Table S2, whose reachable set is a single irreducible class containing the consensus states, with $\tilde{c} > 0$.

C. The epoch chain

Definitions. Assume R1 and $Q = 2$. Call fire k a rotation if $U_k \in \mathcal{M}(Y_{k-1})$ (for $k = 1$ and a tie at time 0, call the first fire a rotation; this convention does not affect long-run averages). Let $\kappa_1 < \kappa_2 < \cdots$ be the indices of rotations. Each κ_m is a stopping time for $(\mathcal{F}_k)$, and $\kappa_m < \infty$ for all m almost surely, because each fire is a rotation with conditional probability $M(Y_{k-1}) \geq 1/2$ (conditional Borel–Cantelli). The *epoch* A_m is the minority set of Y_{κ_m} (empty for consensus), $a_m = |A_m|$, and $i_m = U_{\kappa_m}$. Cycle m is the time interval $[S_{\kappa_m}, S_{\kappa_{m+1}})$, with $K_m = \kappa_{m+1} - \kappa_m$ fires, length $\tau_m = S_{\kappa_{m+1}} - S_{\kappa_m}$ and reward $\mathcal{Y}_m = \int_{\text{cycle } m} M(X_s)\, ds$. Let $\mathcal{H}_m = \mathcal{F}_{\kappa_m}$.

Lemma S4.7 (Epoch invariant and self-containment)**.** *Assume R1 and R2 ($Q = 2$; equal balls not required). Then for every $m \geq 1$: $A_m \subseteq R_{i_m}$; consequently $A_m \subseteq B_j$ for every $j \in A_m$, and $a_m \leq n - \beta_{i_m}$.*

Proof. By Proposition S4.3(a) applied to the rotation κ_m, the minority after it is $\mathcal{M}(Y_{\kappa_m - 1}) \setminus B_{i_m} \subseteq V \setminus B_{i_m} = R_{i_m}$; if $Y_{\kappa_m - 1}$ is consensus the minority is exactly R_{i_m}. (The case of a tie at time 0 is covered by Proposition S4.3(c), whose computation gives the same conclusion.) For $u, v \in R_{i_m}$, R2 gives $d(u, v) \leq r$, hence $R_{i_m} \subseteq B_j$ for every $j \in R_{i_m}$, and in particular $A_m \subseteq R_{i_m} \subseteq B_j$ for every $j \in A_m$. □

Lemma S4.8 (Cycle structure and Markov property)**.** *Assume R1 and R2, $Q = 2$. For $m \geq 1$, with $u = U_{\kappa_m + 1}$ and $u' = U_{\kappa_m + 2}$:*

(i) if $u \notin A_m$, then $K_m = 1$ and $A_{m+1} = (V \setminus A_m) \setminus B_u$;

(ii) if $u \in A_m$, then $Y_{\kappa_m + 1}$ is consensus, $K_m = 2$, and $A_{m+1} = R_{u'}$.

Consequently, conditionally on $\mathcal{H}_m$, the law of $(A_{m+1}, K_m, \tau_m, \mathcal{Y}_m)$ depends only on A_m, and $(A_m)_{m\geq 1}$ is a time-homogeneous Markov chain on the finite family of subsets of V of size $< n/2$.

Proof. At κ_m the configuration Y_{κ_m} is determined by A_m up to the exchange θ, which commutes with every T_i; since minority sets and rotations are invariant under θ, everything below depends on Y_{κ_m} only through A_m. (i) u is a majority node, so fire $\kappa_m + 1$ is a rotation and the new minority is $\mathcal{M} \setminus B_u$ with $\mathcal{M} = V \setminus A_m$ (Proposition S4.3(a)). (ii) $u \in A_m \subseteq B_u$ by Lemma S4.7, so T_u moves all of A_m and the configuration becomes consensus; fire $\kappa_m + 2$ is then a rotation (every node is a majority node), and from consensus it switches exactly $B_{u'}$, which is a majority by R1, so the new minority is $R_{u'}$. In both cases (A_{m+1}, K_m) is a deterministic function of A_m, u, u', and $\tau_m = G_{\kappa_m+1} + \mathbf{1}\{u \in A_m\} G_{\kappa_m+2}$,

$$\mathcal{Y}_m = \Big(1 - \frac{a_m}{n}\Big) G_{\kappa_m+1} + \mathbf{1}\{u \in A_m\}\, G_{\kappa_m+2}, \tag{S2}$$

because $M = 1 - a_m/n$ during the first interval and $M = 1$ during the second. By the strong Markov property of the i.i.d. sequence $((U_k, G_k))_k$ at the stopping time κ_m, the variables $U_{\kappa_m+1}, U_{\kappa_m+2}, G_{\kappa_m+1}, G_{\kappa_m+2}$ are i.i.d. copies of (U, G) independent of $\mathcal{H}_m$. This proves the claims. The sizes satisfy $a_{m+1} < n/2$ by Proposition S4.3. □

Remark S4.9. An epoch can be empty: if $A_m = R_i$ and $u = i$, then $(V \setminus R_i) \setminus B_i = \emptyset$. Lemma S4.8 covers this case ($u \notin A_m = \emptyset$ always; $K_m = 1$).

D. Exact affine closure

Lemma S4.10 (Overlap identity)**.** *Assume R1, R2. Let A be an epoch state ($A \subseteq B_j$ for all $j \in A$) of size a, and $\mathcal{M} = V \setminus A$. Then*

$$\sum_{i\in\mathcal{M}} |A \cap B_i| = S_A - a^2, \qquad S_A = \sum_{j\in A} \beta_j. \tag{S3}$$

With equal balls, for i uniform in $\mathcal{M}$, $\mathbb{E}[|A \cap B_i|] = a(\beta - a)/(n - a)$.

Proof. By symmetry of d, $\mathbf{1}\{j \in B_i\} = \mathbf{1}\{i \in B_j\}$, so $\sum_{i\in\mathcal{M}} |A \cap B_i| = \sum_{j\in A} |B_j \cap \mathcal{M}| = \sum_{j\in A}(\beta_j - |B_j \cap A|) = \sum_{j\in A}(\beta_j - a)$, where $|B_j \cap A| = a$ by self-containment. □

Lemma S4.11 (Affine conditional mean)**.** *Under R1, R2 and equal balls, for every epoch state A of size a, $\mathbb{E}[a_{m+1} \mid \mathcal{H}_m] = (1 - q)(n - a_m)$ on $\{A_m = A\}$.*

Proof. By Lemma S4.8, with u uniform: if $u \in A$ (probability a/n), $a_{m+1} = |R_{u'}| = n - \beta$; if $u \notin A$, $a_{m+1} = |\mathcal{M}| - |\mathcal{M} \cap B_u| = (n - a) - \beta + |A \cap B_u|$, and u is uniform on $\mathcal{M}$ conditionally on $u \notin A$. By Lemma S4.10,

$$\mathbb{E}[a_{m+1} \mid \mathcal{H}_m] = \frac{a}{n}(n - \beta) + \frac{n - a}{n}\Big(n - a - \beta + \frac{a(\beta - a)}{n - a}\Big) = \frac{(n - \beta)(n - a)}{n}.$$ □

Lemma S4.12 (Stationary epoch mean)**.** *Under the assumptions of Lemma S4.11, every invariant probability π of the epoch chain satisfies $\bar{x} := \mathbb{E}_\pi[a]/n = (1 - q)/(2 - q)$.*

Proof. Invariance gives $\mathbb{E}_\pi[a] = \sum_A \pi(A)\mathbb{E}[a_{m+1} \mid A_m = A] = (1 - q)(n - \mathbb{E}_\pi[a])$, a linear equation with the unique solution stated. □

E. Main theorem

Theorem S4.13 (Exact stationary majority fraction)**.** *Assume $Q = 2$, R1, R2 and equal balls β. For every initial law, the long-run time average of $M(X_t)$ exists almost surely and equals $(2 - q)/(3 - 2q)$. In particular $\mu_{\mathcal{C}}(M) = (2 - q)/(3 - 2q)$ for every closed class $\mathcal{C}$ of the configuration chain.*

Proof. Step 1 (martingale decomposition of rewards). By Lemma S4.8 and Eq. (S2), with u and the two exponential gaps independent of $\mathcal{H}_m$,

$$\mathbb{E}[\mathcal{Y}_m \mid \mathcal{H}_m] = \Big(1 - \frac{a_m}{n}\Big)\frac{1}{n} + \frac{a_m}{n} \cdot \frac{1}{n} = \frac{1}{n}, \qquad \mathbb{E}[\tau_m \mid \mathcal{H}_m] = \frac{1}{n}\Big(1 + \frac{a_m}{n}\Big).$$

$\mathcal{Y}_m$ and τ_m are $\mathcal{H}_{m+1}$-measurable and bounded by $G_{\kappa_m+1}+G_{\kappa_m+2}$, whose second moment is $6/n^2$. Hence $\mathcal{Y}_m-1/n$ and $\tau_m-(1+a_m/n)/n$ are martingale differences with respect to $(\mathcal{H}_{m+1})$ with bounded second moments, and by the martingale strong law, almost surely,

$$\frac{1}{N}\sum_{m=1}^{N}\mathcal{Y}_m\to\frac{1}{n},\qquad \frac{1}{N}\sum_{m=1}^{N}\tau_m-\frac{1}{n}\Big(1+\frac{1}{N}\sum_{m=1}^{N}\frac{a_m}{n}\Big)\to 0.$$

Step 2 (ergodic average of the epoch chain). The epoch chain is a finite Markov chain (Lemma S4.8); almost surely it enters a closed class c and $N^{-1}\sum_{m\le N}a_m/n\to\mathbb{E}_{\pi_c}[a]/n=\bar{x}$, where π_c is the stationary law on c and $\bar{x}=(1-q)/(2-q)$ by Lemma S4.12, *the same for every class. Step 3 (ratio and boundary terms).* Let $N(t)=\max\{m:S_{\kappa_m}\le t\}$. Then $\int_0^t M=\int_0^{S_{\kappa_1}}M+\sum_{m<N(t)}\mathcal{Y}_m+\epsilon_t$ with $0\le\epsilon_t\le\tau_{N(t)}$. The first term is finite, and $\tau_N/N\to 0$ almost surely (it is the difference of two sequences with the same Cesàro limit), so dividing by $t\in[S_{\kappa_{N(t)}},S_{\kappa_{N(t)+1}})$ gives

$$\lim_{t\to\infty}\frac{1}{t}\int_0^t M(X_s)\,ds=\frac{1/n}{(1+\bar{x})/n}=\frac{1}{1+\bar{x}}=\frac{2-q}{3-2q}\quad\text{a.s.}$$

The limit is the same on every closed class, so it does not depend on the initial law; comparing with Lemma S4.1 gives $\mu_{\mathcal{C}}(M)=(2-q)/(3-2q)$ for every $\mathcal{C}$. □

Remark S4.14 (What is and is not used). The proof uses neither independence of cycles nor uniqueness of the stationary law. Equal balls enter only through Lemma S4.11; R2 enters only through self-containment (Lemma S4.7); R1 enters through Proposition S4.3 and the fact that a fire from consensus creates a minority.

F. Cycle law and further closed forms

Theorem S4.15 (Two-point cycle law). *Under the assumptions of Theorem S4.13, let the epoch chain be started from an invariant law π (on any class). Then for a cycle, $K\in\{1,2\}$ with $P(K=2)=\bar{x}=(1-q)/(2-q)$, $\mathbb{E}[K]=1+\bar{x}=1/\mathbb{E}[M]$, $\mathbb{E}[T]=(1+\bar{x})/n$ and $\mathrm{CV}_T^2=(7-8q+2q^2)/(3-2q)^2$. From any initial law, the empirical frequency of $\{K=2\}$ and the empirical mean of K over the first N cycles converge almost surely to $\bar{x}$ and $1+\bar{x}$.*

Proof. By Lemma S4.8, $K=1+B$ with $B=\mathbf{1}\{u\in A_m\}$ and $P(B=1\mid\mathcal{H}_m)=a_m/n$, so under π, $P(B=1)=\bar{x}$. $T=G_1+BG_2$ with G_1,G_2 i.i.d. exponential of mean $1/n$ and independent of (A_m,B). Hence $\mathbb{E}[T]=(1+\bar{x})/n$ and, conditioning on B, $\mathrm{Var}(T)=\mathbb{E}[(1+B)]/n^2+\mathrm{Var}(B)/n^2=(1+2\bar{x}-\bar{x}^2)/n^2$. With $\bar{x}=(1-q)/(2-q)$: $1+\bar{x}=(3-2q)/(2-q)$ and $1+2\bar{x}-\bar{x}^2=(7-8q+2q^2)/(2-q)^2$. The empirical statements follow from the martingale strong law applied to B_m-a_m/n and Step 2 of the proof of Theorem S4.13. □

Corollary S4.16 (Further closed forms). *Under the assumptions of Theorem S4.13, almost surely: the long-run time average of $|m|=2M-1$ is $1/(3-2q)$; the long-run rotation rate is $n(2-q)/(3-2q)$; the long-run rate of one-way erasure transitions is $n(1-q)/(3-2q)$.*

Proof. Theorem S4.13, Corollary S4.4(b) and Remark S4.6. □

Corollary S4.17 (Threshold location of the closed form). *$f(q)=(2-q)/(3-2q)$ is strictly increasing, $f'(q)=(3-2q)^{-2}$, $f(1/2)=3/4$, $f(2/3)=4/5$. A crossing of $M=3/4$ predicted by f would lie at $q=1/2$, outside the exact regime; this concerns the closed form, not the dynamics there. For two-dimensional lattices R1∧R2 was verified by breadth-first search at the sizes used ($L=12$, $r=4$, $\beta=111$; $L=30$, $r=10$, $\beta=699$).*

G. Unequal balls

Proposition S4.18 (Unequal balls). *Assume $Q=2$, R1, R2, and $\beta_i<n$ for some i. Let $\bar{\beta}=n^{-1}\sum_i\beta_i$, $\bar{q}=\bar{\beta}/n$, $q_{\min}=\min_i\beta_i/n$, $q_{\max}=\max_i\beta_i/n$. For each closed class c of the epoch chain with invariant law π_c, let $q_c^*=\mathbb{E}_{\pi_c}[S_A]/(n\mathbb{E}_{\pi_c}[a])$. Then $q_c^*\in[q_{\min},q_{\max}]$; on the event that the epoch chain enters c, the long-run time average of M is*

$$g(q_c^*),\qquad g(s)=\frac{2+\bar{q}-2s}{3-2s};\tag{S4}$$

and for every initial law the long-run time average lies in $[g(q_{\max}),g(q_{\min})]$ almost surely.

Proof. Conditional mean. Lemmas S4.7 and S4.8 do not use equal balls. On $\{A_m = A\}$: if $u \in A$ (probability a/n), $a_{m+1} = n - \beta_{u'}$ with u' uniform, of mean $n - \bar{\beta}$; if $u \notin A$, u is uniform on $\mathcal{M} = V \setminus A$ and $a_{m+1} = (n-a) - \beta_u + |A \cap B_u|$. Using $\sum_{i \in \mathcal{M}} \beta_i = n\bar{\beta} - S_A$ and Eq. (S3),

$$\mathbb{E}[a_{m+1} \mid u \notin A, A] = n - a + \frac{2S_A - n\bar{\beta} - a^2}{n - a},$$

and weighting,

$$\mathbb{E}[a_{m+1} \mid \mathcal{H}_m] = n - \bar{\beta} - a\Big(1 + \frac{\bar{\beta}}{n}\Big) + \frac{2S_A}{n}.$$

Stationary mean. Under π_c, $\mathbb{E}[a](2+\bar{q}) = n(1-\bar{q}) + 2q_c^* \mathbb{E}[a]$, so $\bar{x}_c = (1-\bar{q})/(2+\bar{q}-2q_c^*)$. $\mathbb{E}_{\pi_c}[a] > 0$: otherwise π_c would be the point mass at $A = \emptyset$, but from $\emptyset$ the next epoch is R_u with u uniform (Lemma S4.8(i)), which is nonempty with probability at least $1/n$ because $\beta_i < n$ for some i; this contradicts invariance. Since $nq_{\min} \le \beta_j \le nq_{\max}$, $a\, nq_{\min} \le S_A \le a\, nq_{\max}$, which gives the range of q_c^*. *Renewal–reward.* Eq. (S2) and Steps 1–3 of the proof of Theorem S4.13 hold verbatim with $\bar{x}$ replaced by $\bar{x}_c$, giving the time average $1/(1+\bar{x}_c) = g(q_c^*)$. *Bounds.* $g'(s) = -2(1-\bar{q})/(3-2s)^2 \le 0$, so $g(q_{\max}) \le g(q_c^*) \le g(q_{\min})$ for every class. □

Remark S4.19. *Numerical verification* (`scripts/hetero_balls_check.py`). We generated rings with $9 \le n \le 14$ plus up to five random chords, radius $r \in \{2, 3, 4\}$, kept the ball systems with unequal sizes, no full ball, and R1∧R2 verified by breadth-first search, and removed duplicate ball systems; 60 distinct systems remained. For each, the stationary law of the configuration chain on the states reachable from consensus was obtained by a sparse direct solve (L^1 residual $\le 3.1 \times 10^{-15}$). The epoch law was computed independently as the rotation flux into each state, $\propto \sum_x \mu(x) n^{-1} \#\{i \in \mathcal{M}(x) : T_i(x) = y\}$, and q^* from it. Results: $\max |\mathbb{E}[M] - g(q^*)| = 1.2 \times 10^{-15}$; the bounds hold in all 60 cases; $\max |(\text{rotation fraction}) - \mathbb{E}[M]| = 2.0 \times 10^{-14}$ (Corollary S4.4); $q^* < \bar{q}$ in all 60 cases, and $\mathbb{E}[M] - (2-\bar{q})/(3-2\bar{q}) \in [0.0008, 0.0127]$. No proof of $q^* \le \bar{q}$ is known.

S5. CONTAINMENT THEORY

Proposition S5.1 (c_G on vertex-transitive geometries)**.** *$c_G = \mathbb{E}_{i,j} \mathbf{1}\{R_i \subseteq B_j\}$ with i, j independent and uniform. On a vertex-transitive graph, $c_G = n^{-1} |\{j : R_i \subseteq B_j\}|$ for every i. In general c_G is not determined by ball sizes and must be computed by enumeration.*

Proof. The first identity is the definition. If ϕ is an automorphism with $\phi(i) = i'$, then $\phi(B_j) = B_{\phi(j)}$ and $\phi(R_i) = R_{i'}$, so $j \mapsto \phi(j)$ is a bijection between $\{j : R_i \subseteq B_j\}$ and $\{j : R_{i'} \subseteq B_j\}$; the count is independent of i. □

Proposition S5.2 (Necessary condition)**.** *If $R_i \subseteq B_j$ then* $\operatorname{diam}(R_i) \le 2r$.

Proof. $d(u, v) \le d(u, j) + d(j, v) \le 2r$ for $u, v \in R_i \subseteq B_j$. □

Proposition S5.3 (R2 implies one-step extinction)**.** *Under R2, $\{j \in R_i : R_i \subseteq B_j\} = R_i$ for every i, hence $\tilde{c} = 1$. Uniform balls are not required.*

Proof. For $j, v \in R_i$, $d(j, v) \le \operatorname{diam}(R_i) \le r$, so $v \in B_j$. □

Proposition S5.4 (Intersection form, equal balls)**.** *$R_i \subseteq B_j \iff B_i \cup B_j = V \iff |B_i \cap B_j| = 2\beta - n$; in particular $c_G = 0$ if $2\beta < n$. Hence c_G is computed exactly by counting pairs with* $\operatorname{popcount}(B_i \,\&\, B_j) = 2\beta - n$.

Proof. $R_i \subseteq B_j$ iff $V \setminus B_i \subseteq B_j$ iff $B_i \cup B_j = V$; by inclusion–exclusion $|B_i \cup B_j| = 2\beta - |B_i \cap B_j|$, which equals n iff $|B_i \cap B_j| = 2\beta - n$. If $2\beta < n$ this is impossible. □

Proposition S5.5 (Admissible centres on the ring)**.** *On the ring with $\beta = 4r + 1 < n$, fix i and write $j = i + s$ with $s \in \mathbb{Z}_n$. Then $R_i \subseteq B_j$ iff s has a representative $t \in \mathbb{Z}$ with*

$$n - 4r - 1 \le t \le 4r + 1. \tag{S5}$$

The number of such s is $\max(0, 2\beta - n + 1)$.

Proof. Lift to $\mathbb{Z}$ with $i = 0$: R_0 is the integer interval $I = [2r+1,\, n-2r-1]$ of length $\ell = n - \beta \geq 1$, and B_s is the union of the disjoint integer intervals $J_k = [s - 2r + kn,\, s + 2r + kn]$, $k \in \mathbb{Z}$, separated by gaps of $\ell \geq 1$ integers. An interval of consecutive integers contained in such a union lies in a single J_k; with $t = s + kn$, $I \subseteq [t - 2r, t + 2r]$ is Eq. (S5). The admissible t form an integer interval of length $L = 8r + 3 - n = 2\beta - n + 1$ when positive; since $L \leq n - 1 < n$ (because $\beta < n$), distinct admissible t are distinct modulo n, so there are exactly $\max(0, L)$ admissible s. □

Proposition S5.6 (c_G on the ring)**.** *With $\beta = \min(n, 4r+1)$, $c_G = \min\{1, \max(0, 2\beta - n + 1)/n\}$; at $2\beta = n$, $c_G = 1/n$.*

Proof. If $\beta = n$, residuals are empty and $c_G = 1$. Otherwise the ring is vertex-transitive, so by Propositions S5.1 and S5.5, $c_G = \max(0, 2\beta - n + 1)/n$, which is < 1. At $2\beta = n$ the count is 1. □

Proposition S5.7 ($\tilde{c}$ on the ring)**.** $\tilde{c} = 0$ *if* $2\beta < n$; $\tilde{c} = (2\beta - n + 1)/(n - \beta)$ *if* $2\beta \geq n$ *and* $3\beta < 2n - 1$ *(equal to* $1/\beta$ *at* $2\beta = n$*);* $\tilde{c} = 1$ *if* $3\beta \geq 2n - 1$*. It jumps from* 0 *to* $1/\beta$ *at* $2\beta = n$ *and is continuous at* $3\beta = 2n - 1$*.*

Proof. If $\beta = n$, $\tilde{c} = 1$ by convention. Let $\beta = 4r + 1 < n$. By symmetry every i contributes the same ratio, so $\tilde{c} = |\{j \in R_i : R_i \subseteq B_j\}|/(n - \beta)$. In the lift of Proposition S5.5, $j \in R_0$ means $t \in [2r+1, n-2r-1]$ for the representative in $[0, n)$, and every admissible t of Eq. (S5) lies in $[0, n)$ when $2\beta \geq n$ (as $n - 4r - 1 \geq 0$ and $4r + 1 < n$). So we count integers in $[\max(n - 4r - 1, 2r + 1),\, \min(4r + 1, n - 2r - 1)]$. If $2\beta < n$ there are none (Proposition S5.5). If $2\beta \geq n$ and $n \geq 6r + 2$, then $n - 4r - 1 \geq 2r + 1$ and $4r + 1 \leq n - 2r - 1$, so the count is $2\beta - n + 1$. If $n \leq 6r + 2$, the interval of admissible t contains $[2r + 1, n - 2r - 1]$ and the count is $n - \beta$, i.e., $\tilde{c} = 1$. With $\beta = 4r + 1$, $n \geq 6r + 2$ is $3\beta \leq 2n - 1$; at equality $(2\beta - n + 1)/(n - \beta) = 1$, which proves continuity and that the middle formula may be used for $3\beta < 2n - 1$. At $2\beta = n$ the count is 1 and $n - \beta = \beta$. □

Proposition S5.8 (Three regions on the ring)**.** *Region I:* $3\beta \geq 2n - 1$ *(R1 and R2,* $\tilde{c} = 1$*). Region II:* $n/2 < \beta < (2n - 1)/3$ *(R1 only,* $0 < \tilde{c} < 1$*). Boundary:* $2\beta = n$ *(*$c_G = 1/n$*,* $\tilde{c} = 1/\beta$*). Region III:* $2\beta < n$ *(*$\tilde{c} = 0$*).*

Proof. Combine Propositions S3.2, S5.6 and S5.7. In Region II, $2\beta - n + 1 \geq 2$ and $2\beta - n + 1 < n - \beta$, so $0 < \tilde{c} < 1$. □

Proposition S5.9 (Newborn identity)**.** *Let $Q = 2$. From consensus, a fire at i creates the minority R_i. Conditionally on the next fire being at a minority node, that node is uniform in R_i, and the minority is erased if and only if $R_i \subseteq B_j$ for the firing node j. Averaging over i (uniform) with weights $|R_i|$ gives $\tilde{c}$.*

Proof. The firing node is uniform on V and independent of the past, hence uniform on R_i given that it lies there. A fire at $j \in R_i$ moves $R_i \cap B_j$; the minority disappears iff this is all of R_i. The probability that the newborn is R_i and the next minority fire erases it, normalized by the probability that the next fire is a minority fire, is $\sum_i |\{j \in R_i : R_i \subseteq B_j\}| / \sum_i |R_i| = \tilde{c}$. □

Validation. Propositions S5.6–S5.7 agree exactly (rational arithmetic, explicit set inclusion) with exhaustive enumeration for every $n \in [12, 40]$ and every radius, 192 cases, including the boundary case $(n, r) = (18, 2)$ with $c_G = 1/18$, $\tilde{c} = 1/9$ (`tests/independent/containment_bruteforce.py`). The Monte Carlo estimate of the newborn erasure probability agrees with $\tilde{c}$ within binomial error on native ring balls, e.g., $n = 15$, $r = 2$: 29 489 trials, 0.6664 ± 0.0027 against $2/3$; $n = 25$, $r = 3$: 19 535 trials, 0.1666 ± 0.0027 against $1/6$; and in all 18 cells with $0 < \tilde{c} < 1$ of Protocol S7.1 (largest $|z| = 1.85$).

S6. EXACT SOLVERS, SIMULATIONS AND ANNEALED COMPUTATIONS

Proposition S6.1 (Exact Region-II values; computer-assisted)**.** *For the six odd Region-II ring cases of Table S2, the stationary majority fraction equals the rational number listed there, independently of the initial law.*

Proof. (1) *Exact lumping.* Every automorphism ϕ of the ring (the dihedral group D_n) satisfies $\phi(B_i) = B_{\phi(i)}$, hence $\phi \circ T_i = T_{\phi(i)} \circ \phi$; together with $\theta \circ T_i = T_i \circ \theta$ this shows that the jump chain $P = n^{-1} \sum_i T_i$ commutes with the action of $D_n \times \{1, \theta\}$, so it lumps exactly onto orbits, and M is constant on orbits. (2) *Uniqueness.* The set of configurations reachable from consensus was enumerated and found to be a single closed communicating class (Protocol S6.2); since consensus is reachable from every configuration with a strict majority (a majority fire is a rotation and, by repeated minority fires, the chain reaches consensus with positive probability), every initial law leads into this class, and its stationary law μ is unique. (3) *Exact solution.* The lumped linear system $\mu(P - I) = 0$, $\sum \mu = 1$ was solved in exact rational arithmetic (FLINT, Dixon's algorithm), and the solution was verified by exact substitution. Then $\mu(M)$ is the listed rational, and Lemma S4.1 identifies it with the long-run time average. □

TABLE S1. Annealed version: exact stationary $\mathbb{E}[M]_{\text{ann}}$ for $\beta = \text{round}(qn)$ and the gap $f(\beta/n) - \mathbb{E}[M]_{\text{ann}}$ to the quenched exact value. Last row: large-n random map of Heuristic S6.4 (4×10^6 iterations, batch-means standard error).

	$q = 0.7$ ($f = 0.81250$)		$q = 0.8$ ($f = 0.85714$)		$q = 0.9$ ($f = 0.91667$)	
n	$\mathbb{E}[M]_{\text{ann}}$	gap	$\mathbb{E}[M]_{\text{ann}}$	gap	$\mathbb{E}[M]_{\text{ann}}$	gap
100	0.79524	0.01726	0.85031	0.00683	0.91550	0.00116
200	0.79499	0.01751	0.85016	0.00699	0.91544	0.00122
400	0.79486	0.01764	0.85008	0.00706	0.91541	0.00125
800	0.79480	0.01770	0.85004	0.00710	0.91540	0.00127
1600	0.79477	0.01773	0.85002	0.00712	0.91539	0.00127
3200	0.79475	0.01775	—	—	—	—
map	0.79472(2)	—	0.84999(2)	—	0.91539(1)	—

Protocol S6.2 (Exact solvers and simulation checks)**.** *(i) Reduced exact solver* (`tests/gvm_validation_suite.py`, task T5). States are orbits of reachable minority sets under the dihedral group and state exchange; lazy power iteration from the consensus orbit, L^1 residual $\leq 1.4 \times 10^{-14}$ (Table S2). *(ii) Independent cross-check* (`tests/independent/exact_chain_independent.py`). Full configurations, no symmetry reduction, transitions generated literally from the update rule, irreducibility of the reachable set checked; sparse direct solve ($n \leq 21$) or lazy power iteration to L^1 change $< 10^{-16}$ ($n = 23, 25$). The lumped chain is also solved in exact rational arithmetic (FLINT, Dixon's algorithm) and verified by exact substitution (Table S3). *(iii) Simulations* (`tests/independent/sim_independent.py`). Independent continuous-time simulator, 10 replicas of 2×10^6 fires per case. Region II, native ring balls, the six cases of Table S2: largest $|M_{\text{sim}} - M_{\text{exact}}| = 4.6 \times 10^{-5}$, largest $|z| = 1.6$. Region I, native balls with R1 and R2 verified by breadth-first search (ring $n = 400$, $r = 75$; $n = 401$, $r = 70$; 2D $L = 12$, $r = 4$; $L = 30$, $r = 10$): largest $|M_{\text{sim}} - f(q)| = 2.1 \times 10^{-5}$ (largest $|z| = 0.8$); cycle law (Theorem S4.15): largest $|z| = 1.9$ for $\mathbb{E}[K]$ and for CV_T^2.

Protocol S6.3 (Extended Region-II solutions and annealed version)**.** *(i) Quenched chain* (`scripts/region2_exact.c`). Written independently of Protocol S6.2. States: configurations reachable from consensus by breadth-first search, identified under state exchange only. Since every node fires at rate one, the stationary law equals that of the jump chain $P = n^{-1} \sum_i T_i$; power iteration of $(I + P)/2$ until the L^1 change is below 10^{-12}, then 3000 further iterations; residual $\|\mu P - \mu\|_1 < 5 \times 10^{-13}$ in every case. It reproduces the six exact rationals to 10^{-12}. *(ii) Annealed version* (`scripts/annealed_baseline.py`). At every fire of i the interaction set is $\{i\}$ plus $\beta - 1$ nodes drawn uniformly without replacement from $V \setminus \{i\}$. If i has a state shared by m nodes, $1 + h$ nodes switch, with h hypergeometric (population $n - 1$, $m - 1$ successes, $\beta - 1$ draws). The chain on the count n_1 is solved by a direct linear solve (residual $< 2 \times 10^{-13}$ up to $n = 400$, $< 10^{-12}$ up to $n = 3200$). *(iii) Independent re-implementation* (`scripts/annealed_size_scan.py`). A separate implementation of (ii) reproduces all 17 annealed values of main-text Table II to 5×10^{-7} (the table's rounding) and the four $n = 400$ values quoted in the main text, and produces Table S1 and the $n = 400$ curve of main-text Fig. 2 (`scripts/fig_quenched_annealed.py`).

At $n = 400$ the annealed value is below $f(\beta/n)$ for every $\beta \leq n - 2$ in Region I of the ring ($267 \leq \beta \leq 398$), with gaps from 1.2×10^{-7} ($\beta = 398$) to 0.022; at $\beta = n - 1$ the gap is zero to machine precision, since each interaction set then misses a single node.

Heuristic S6.4 (Large-n annealed map)**.** *For $q > 1/2$ and $n \to \infty$ at fixed q, hypergeometric concentration suggests that the minority fraction x of the annealed version evolves, per fire, by the random map $x \to (1-q)x$ with probability x (minority fire) and $x \to (1-q)(1-x)$ with probability $1 - x$ (majority fire, always a rotation for $q > 1/2$). No proof of convergence is attempted; the stationary mean of $1 - x$ under this map agrees with the exact chain at the largest sizes computed (Table S1). The quenched counterparts are exact: $x \to 0$ for a minority fire (Lemma S4.7) and conditional mean $(1 - q)(1 - 2x)/(1 - x)$ for a majority fire (Lemma S4.10).*

S7. EXPLORATORY REGRESSION TEST OF THE ONE-STEP CONTAINMENT HYPOTHESIS

Protocol S7.1 (Exploratory regression test)**.** *Design.* Families: ring (cycle); two-dimensional torus with king (eight-neighbor) adjacency; configuration model with degree sequence $k = 6$, drawn as a multigraph and projected to a simple graph (self-loops removed, multi-edges merged), hence not exactly 6-regular; Watts–Strogatz with $k = 6$, $p = 0.1$. Sizes $n \in \{100, 144, 400\}$; coverages $q \in \{0.55, 0.60, \ldots, 0.80\}$. One graph per (family, n), seed 42; three runs (seeds 1, 2, 3) per (family, n, q): 216 runs.

TABLE S2. Exact chain in Region II of the ring, odd $n \leq 25$. Reachable: minority sets reachable from consensus (one irreducible class in every case); orbits: classes under the dihedral group and state exchange.

n	r	β	q	c_G	$\tilde{c}$	$\mathbb{E}[M]$	$f(q)$	reachable	orbits
9	1	5	0.5556	2/9	1/2	43/57	13/17	73	8
15	2	9	0.6000	4/15	2/3	65/84	7/9	481	24
17	2	9	0.5294	2/17	1/4	149/201	25/33	2177	80
21	3	13	0.6190	2/7	3/4	265/339	29/37	2689	80
23	3	13	0.5652	4/23	2/5	72/95	33/43	11777	288
25	3	13	0.5200	2/25	1/6	319/433	37/49	51201	1088

TABLE S3. Independent cross-check of Table S2: full configurations, no symmetry reduction. Config.: reachable configurations. Last column: floating-point value minus the exact rational.

n	r	config.	$\mathbb{E}[M]$ (floating point)	L^1 residual	difference
9	1	146	0.7543859649	1.4×10^{-15}	-1.1×10^{-16}
15	2	962	0.7738095238	1.3×10^{-14}	1.1×10^{-16}
17	2	4354	0.7412935323	2.7×10^{-14}	1.2×10^{-14}
21	3	5378	0.7817109145	4.1×10^{-14}	5.7×10^{-15}
23	3	23554	0.7578947368	9.6×10^{-16}	-2.1×10^{-15}
25	3	102402	0.7367205543	9.4×10^{-15}	2.1×10^{-15}

Interaction sets. $\beta = \mathrm{round}(qn)$; B_i consists of the β nodes nearest to i under breadth-first layering, ties in the last layer broken at random with the run seed. The three runs of a cell therefore share the graph but not necessarily the interaction sets; $\tilde{c}$ differs between runs in 15 of the 72 cells. Truncated sets need not satisfy $j \in B_i \iff i \in B_j$, so this is a modified interaction-set model, not the native-ball model of the theorems.

Measurement. $Q = 2$; 2×10^5 fires per run, 5×10^4 discarded; M and its Monte Carlo standard error from 75 batch means of 2000 fires; c_G and $\tilde{c}$ by exact enumeration on the truncated sets. Data: `data/processed/measurements_v2.csv` (one row per run).

Model. Weighted least squares, weights $1/\mathrm{se}^2$: $M \sim q + C(\mathrm{family}) + C(n)$ (reduced) and $M \sim q + \tilde{c} + C(\mathrm{family}) + C(n)$ (full); γ is the coefficient of $\tilde{c}$.

Family-effect norm. $N = [\sum_{f<g}(\alpha_f - \alpha_g)^2]^{1/2}$ over all six pairs of family effects, invariant under the choice of reference; reduction $(N_{\mathrm{red}} - N_{\mathrm{full}})/N_{\mathrm{red}}$.

Primary bootstrap. Within each family, its three (family, n) clusters are resampled with replacement; within each selected cluster, the three runs are resampled inside each of the six coverage strata. $B = 2000$, seed 9999, percentile intervals.

Acceptance criteria (fixed before analysis). (a) $\gamma > 0$ with interval excluding zero; (b) reduction ≥ 0.30.

Result. $\hat{\gamma} = -0.0196$ [95% interval $(-0.0222, -0.0169)$]; reduction -0.98 $[(-1.13, -0.85)]$; fraction of replicates with reduction ≥ 0.30: 0.000. Neither criterion is met.

Status. Exploratory: the resampling scheme and the family statistic were chosen after a first analysis (Note S7.2).

Limitations. (1) $\tilde{c}$ describes only the first minority fire after a minority is created from consensus. (2) Within families, $\tilde{c}$ varies almost only in the ring family (96.6% of the within-family sum of squares; it is identically zero in all 2D cells and at most 0.241 (WS) and 0.036 (configuration)), so γ is weakly identified. (3) Only 12 clusters, each a single graph realization. (4) Coverage enters linearly in the primary model. (5) A failed criterion does not identify an alternative mechanism.

Note S7.2 (Analysis record and sensitivity analyses)**.** Only the acceptance criteria were fixed before the data were analyzed. The initial analysis resampled runs independently and measured family separation relative to the ring reference; an intermediate analysis used a two-level cluster bootstrap and the norm of the three contrasts among the non-ring families. Both were replaced, for the reasons below; all items are reproduced by `src/analysis/reanalysis_v2.py`. *1. Point estimate.* Full-sample estimates: -0.0191 without $C(n)$, -0.0196 with $C(n)$; the intermediate analysis reported the bootstrap mean (-0.0165). All point estimates quoted are full-sample fits. *2. Family statistic.* The three-contrast norm omits all contrasts involving the ring family, the only one in which $\tilde{c}$ varies appreciably, and is not reference-invariant. With all six contrasts the reduction is -0.98; the three-contrast statistic gives -0.047 $[(-0.167, +0.018)]$. *3. Bootstrap.* The unstratified two-level scheme omitted at least one family in 61 of 500 replicates

TABLE S4. Regression test, all specifications and resampling schemes. Point: full-sample estimate. Intervals: bootstrap percentile 95%. Reduction: six-contrast family-norm reduction (negative means increased separation). Missing: replicates lacking at least one family. †: specification of the intermediate analysis (no $C(n)$, $B = 500$).

Specification	Bootstrap	B	$\hat{\gamma}$	γ 95%	reduction 95%	missing
lin. q, no $C(n)^{\dagger}$	unstratified cluster	500	-0.0191	$(-0.0230, +0.0045)$	$(-1.269, -0.003)$	61
lin. q, no $C(n)^{\dagger}$	runs iid	500	-0.0191	$(-0.0220, -0.0163)$	$(-1.251, -0.737)$	0
primary: lin. q, $C(n)$	unstratified cluster	2000	-0.0196	$(-0.0236, +0.0022)$	$(-1.365, -0.001)$	258
primary: lin. q, $C(n)$	stratified cluster	2000	-0.0196	$(-0.0222, -0.0169)$	$(-1.129, -0.849)$	0
primary: lin. q, $C(n)$	runs iid	2000	-0.0196	$(-0.0223, -0.0170)$	$(-1.249, -0.753)$	0
primary: lin. q, $C(n)$	within cells	2000	-0.0196	$(-0.0200, -0.0192)$	$(-1.021, -0.937)$	0
lin. q, no $C(n)$	stratified cluster	2000	-0.0191	$(-0.0221, -0.0165)$	$(-1.116, -0.821)$	0
categ. q, $C(n)$	stratified cluster	2000	-0.0185	$(-0.0208, -0.0161)$	$(-1.074, -0.802)$	0
(n, q)-cell FE	stratified cluster	2000	-0.0186	$(-0.0215, -0.0159)$	$(-1.119, -0.772)$	0
primary, OLS	—	—	-0.0181	—	point -0.84	—
primary, 72 cell means	—	—	-0.0188	—	point -0.94	—

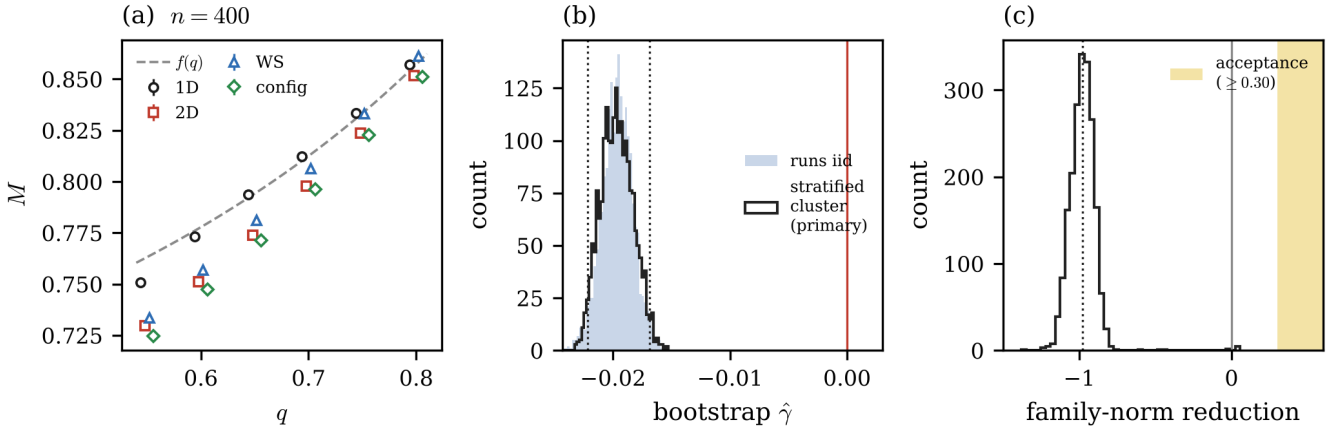


FIG. S1. Exploratory test. (a) Simulated M against q at $n = 400$ for the four families (mean of three runs per cell; standard errors smaller than the markers); $f(q)$ is shown as a comparator, the truncated sets do not satisfy the assumptions of the main-text Theorem 2. (b) Bootstrap distributions of $\hat{\gamma}$: primary family-stratified cluster bootstrap (outline, dotted 95% limits) and independent resampling of runs (filled). (c) Bootstrap distribution of the family-norm reduction under the primary scheme; acceptance region shaded.

(258 of 2000 in the primary specification) and did not preserve coverage strata. All its replicates with $\hat{\gamma} > 0$ lack at least one family; its wide interval [$(-0.0236, +0.0022)$ in the primary specification] reflects replicates in which γ is barely identified. *4. Standard errors.* The between-run standard deviation within a cell divided by the mean batch-means standard error has median 0.832 over the 72 cells, against $\sqrt{\ln 2} \approx 0.833$ expected for three runs with correct standard errors. Fits to the 72 cell means give $\hat{\gamma} = -0.0188$. *5. Dependence.* Resampling runs independently gives $(-0.0223, -0.0170)$; resampling only within cells (Monte Carlo noise alone) gives $(-0.0200, -0.0192)$. *6. Coverage variable.* Using the realized β/n instead of the nominal q (difference $\leq 0.5/n$) gives $\hat{\gamma} = -0.0201$.

Interpretation. Once family fixed effects are included, γ is identified by within-family variation of $\tilde{c}$, almost all of it in the ring family, where $\tilde{c}$ rises between $q = 0.55$ and 0.65 and is then constant at 1 while the advantage of the ring family over the others shrinks. The negative coefficient records this co-variation; it is evidence against the mediation hypothesis in this design, not evidence that one-step containment lowers M. In Region I the ring truncated sets (contiguous arcs) give M within 2×10^{-4} of $f(\beta/n)$ at $q \geq 0.70$; the other families lie below f, which we do not explain.

S8. CONJECTURES, OPEN PROBLEMS AND LIMITATIONS

Conjecture S8.1. *In Region II of the ring, $\mathbb{E}[M] \le f(q)$ for all n and admissible r.*

Evidence. Exact rationals for the six odd $n \le 25$ (Table S2) and floating-point solutions of all 17 Region-II cases with $n \le 33$ (main-text Table II), with deviations from -0.0009 to -0.0195, increasing monotonically with q. No coupling or monotonicity proof is known.

Open problem S8.2 (Unequal balls)**.** *Prove or refute $q^* \le \bar{q}$ under R1∧R2 (Remark S4.19).*

Open problem S8.3 (Fluctuations)**.** *The variance of M is not fixed by the affine closure, because it involves pairwise overlaps $|B_j \cap B_k|$ of balls.*

Open problem S8.4 (Family dependence)**.** *The difference between the configuration-model and 2D families in Protocol S7.1 is unexplained; candidate causes include residual ball-size heterogeneity, expansion properties and fractional-radius effects.*

Heuristic S8.5 (Fragmentation)**.** *Below R2, minorities may fragment; no low-dimensional closed Markov description in terms of shape variables is known.*

Limitations. (1) The closed form requires two states, R1, R2 and equal balls; Proposition S4.18 relaxes the last. (2) Regions II and III have no closed-form characterization. (3) Claims S3.4–S3.5 are not used. (4) The annealed size dependence (Table S1) is computed, not proved, and Heuristic S6.4 is a heuristic. (5) The regression test uses one graph per (family, n) and truncated interaction sets.

S9. VALIDATION LEDGER

Numerical checks cited in the main text and their evidence files (paths relative to the repository root [1]):

- Ring containment formulas vs enumeration: exact agreement, 192 cases. `results/validation/containment_bruteforce.json`.
- Exact chain, Region II, six cases, two independent solvers: agree; rationals certified. `results/validation/exact_independent.json`; `validation_results/T5.json`.
- Exact chain, Region I: equals $f(q)$. `validation_results/T1.json`.
- Simulator vs exact chain, Region II: $|\Delta| \le 4.6 \times 10^{-5}$. `results/validation/sim_independent.json`.
- Simulator vs Theorem S4.13, Region I (ring, 2D): $|\Delta| \le 2.1 \times 10^{-5}$. `results/validation/sim_independent.json`.
- Cycle law, $\mathbb{E}[K]$ and CV_T^2: $|z| \le 1.9$. `results/validation/sim_independent.json`.
- Newborn identity: within binomial error. `validation_results/T6.json`.
- Region II, $n \le 33$, and annealed version: all below f; annealed below quenched. `scripts/region2_results.txt`.
- Proposition S4.18, 60 ball systems: $|\Delta| \le 1.2 \times 10^{-15}$. `scripts/hetero_balls_check.py`.
- Annealed size scan ($n \le 3200$) and random map: Table S1. `scripts/annealed_size_scan.py`; `scripts/annealed_random_map.py`.
- Regression test: executed; criteria not met. `results/tables/reanalysis/`.

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[1] P. L. D. Nguyen, Group-flip consensus on hypergraphs: reproducibility package (code and data), Zenodo (2026).